SQL Data Analysis — Report

Project 3 · Data Analytics Internship · Decode Labs

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Position:	Data Analytics Intern	**Mode:**	Remote / Virtual
Internship Period:	27 Sep 2026 – 27 Oct 2026	**Report Date:**	14 Oct 2026
Dataset:	Orders_Clean (1,200 rows)	**Tool Used:**	SQLite (DB Browser)

Objective

Use SQL queries to extract insights from a dataset, per the Project 3 brief: write SELECT queries, use WHERE, ORDER BY and GROUP BY, and perform basic aggregations (COUNT, SUM, AVG). The analysis runs on the same cleaned order data used in Projects 1 and 2, so SQL results can be checked directly against the earlier Excel work.

Setup

The cleaned Orders_Clean sheet from Project 1 was exported from Excel as a CSV file and imported into a SQLite database (Orders_Database.db) using DB Browser for SQLite, with the first row used as column names. The import was verified with SELECT COUNT(*), which returned 1,200 rows, matching the Project 1 row count. Quantity and ItemsInCart were imported as INTEGER, UnitPrice and TotalPrice as REAL, and all other columns as TEXT.

Definition used in this report: "total order value" is TotalPrice summed over every order placed, including Cancelled and Returned orders. It is not realized revenue. Section 3 shows how the results change when those orders are excluded.

1. Query Log

#	Purpose	SQL used	Result
1	Preview the first 10 rows	SELECT, LIMIT	10 rows, 14 columns
2	Count all rows	COUNT	1,200
3	Select specific columns	SELECT, LIMIT	4 columns, 10 rows
4	Filter by product	WHERE	Laptop orders only
5	Count Laptop orders	WHERE, COUNT	173
6	Ten highest-value orders	ORDER BY DESC, LIMIT	10 rows
7	Orders above outlier cutoff (3,330.41)	WHERE, COUNT	8
8	Laptop orders of exactly 5 units	WHERE, AND, ORDER BY	39 rows
9	Orders and average value per product	GROUP BY, COUNT, AVG	7 rows
10	Orders and average value per referral source	GROUP BY, COUNT, AVG	5 rows

#	Purpose	SQL used	Result
11	Orders and total order value per product	GROUP BY, COUNT, SUM	7 rows
12	Products with more than 170 orders	GROUP BY, HAVING	4 rows
13	Order value share per product	GROUP BY, SUM, subquery	7 rows
14	Product summary excluding Cancelled/Returned	WHERE, NOT IN, GROUP BY	7 rows (703 orders)

The full, commented SQL for all 14 queries is in Project3_SQL_Queries.sql.

2. Key Queries and Results

Query 9 — Orders and average order value per product (GROUP BY, COUNT, AVG)

```
SELECT Product,
       COUNT(*) AS Order_Count,
       ROUND(AVG(TotalPrice), 2) AS Avg_TotalPrice
FROM Orders_Clean
GROUP BY Product
ORDER BY Avg_TotalPrice DESC;
```

Product	Order_Count	Avg_TotalPrice
Laptop	173	1,110.56
Chair	178	1,098.99
Printer	181	1,080.73
Monitor	163	1,077.62
Tablet	179	1,042.28
Desk	170	985.06
Phone	156	972.58

Query 10 — Orders and average order value per referral source

ReferralSource	Order_Count	Avg_TotalPrice
Facebook	228	1,098.29
Instagram	259	1,062.88
Email	250	1,047.23
Google	241	1,039.18
Referral	222	1,021.69

Query 11 — Total order value per product (SUM) and Query 13 — share of total

```
SELECT Product,
       ROUND(SUM(TotalPrice), 2) AS Total_Order_Value,
       ROUND(100.0 * SUM(TotalPrice) /
             (SELECT SUM(TotalPrice) FROM Orders_Clean), 2) AS Order_Value_Share_Pct
FROM Orders_Clean
GROUP BY Product
ORDER BY Total_Order_Value DESC;
```

Product	Order_Count	Total_Order_Value	Order_Value_Share_Pct
Chair	178	195,620.11	15.47
Printer	181	195,612.61	15.47
Laptop	173	192,126.56	15.19
Tablet	179	186,568.95	14.75
Monitor	163	175,651.41	13.89
Desk	170	167,459.93	13.24
Phone	156	151,722.39	12.00

Order counts come from Query 11; shares are from Query 13 (they sum to 100.01 because of rounding).

Query 12 — Filtering groups with HAVING

```
SELECT Product,
       COUNT(*) AS Order_Count
FROM Orders_Clean
GROUP BY Product
HAVING COUNT(*) > 170
ORDER BY Order_Count DESC;
```

Product	Order_Count
Printer	181
Tablet	179
Chair	178
Laptop	173

HAVING filters whole groups after aggregation, whereas WHERE filters individual rows before grouping. Desk has exactly 170 orders, so it is correctly excluded by the strict > 170 condition.

Query 14 — Product summary excluding Cancelled and Returned orders (WHERE, NOT IN)

```
SELECT Product,
       COUNT(*) AS Order_Count,
       ROUND(SUM(TotalPrice), 2) AS Total_Order_Value,
       ROUND(AVG(TotalPrice), 2) AS Avg_TotalPrice
FROM Orders_Clean
WHERE OrderStatus NOT IN ('Cancelled', 'Returned')
GROUP BY Product
ORDER BY Total_Order_Value DESC;
```

Product	Order_Count	Total_Order_Value	Avg_TotalPrice
Printer	108	120,998.36	1,120.36
Chair	105	120,320.97	1,145.91
Laptop	99	108,710.54	1,098.09
Tablet	102	103,413.13	1,013.85
Desk	103	99,040.75	961.56
Monitor	92	98,069.37	1,065.97
Phone	94	94,534.93	1,005.69

These 7 rows cover 703 orders with a combined order value of 745,088.05.

3. Effect of Excluding Cancelled and Returned Orders

Cancelled (250) and Returned (247) orders make up 497 of the 1,200 orders (41.4%). Total order value therefore includes value that may never have become revenue. Comparing Query 11 and Query 9 (all 1,200 orders) with Query 14 (703 orders) shows that several rankings depend on which orders are counted:

Measure	All 1,200 orders	Excluding Cancelled/Returned
Largest total order value	Chair (195,620.11)	Printer (120,998.36)
Highest average order value	Laptop (1,110.56)	Chair (1,145.91)
Lowest average order value	Phone (972.58)	Desk (961.56)
Smallest total order value	Phone (151,722.39)	Phone (94,534.93)

Only one conclusion holds in both views: Phone has the smallest total order value. Statements about which product is best or worst on average order value should be read as applying to all orders placed.

4. Cross-Validation Against Earlier Excel Work

Check	SQL result	Excel result (Projects 1-2)	Match
Total rows	1,200	1,200 (Project 1)	Yes
Laptop order count	173	173 (Product PivotTable)	Yes
Orders above 3,330.41	8	8 (IQR outlier count)	Yes

Check	SQL result	Excel result (Projects 1-2)	Match
Product counts and averages	Query 9	Product PivotTable	Yes
Referral counts and averages	Query 10	Referral PivotTable	Yes
Sum of product order values	1,264,761.96	1,200 x 1,053.9683 = 1,264,761.96	Yes

Queries 8 and 14 have no equivalent earlier result to compare against.

5. Key Findings

1. Counting all 1,200 orders placed, Laptop has the highest average order value (1,110.56) but ranks third in total order value (192,126.56), because Chair (178) and Printer (181) have more orders.

2. Chair and Printer are effectively tied for the highest total order value: 195,620.11 versus 195,612.61, a gap of 7.50. Each is about 15.47% of the total.

3. 41.4% of orders (497 of 1,200) were Cancelled or Returned. Excluding them, Printer ranks first (120,998.36), Chair has the highest average order value (1,145.91), and Laptop's lead on average order value disappears. Rankings depend on which orders are counted.

4. Phone has the smallest total order value in both views: 151,722.39 (12.00%) across all orders and 94,534.93 (12.69%) excluding Cancelled and Returned. Its lowest average order value holds only when all orders are counted.

5. Among referral channels (all orders), Facebook has the highest average order value (1,098.29) while Instagram brings the most orders (259). This was not re-checked with Cancelled and Returned orders excluded.

Notes

The dataset does not state a currency, so monetary values are shown as plain numbers. All queries were run in DB Browser for SQLite, one at a time, against the Orders_Clean table.

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